

Alan Amadeus Davila

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Education

Texas A&M University — College Station, TX

B.S. in Electrical & Computer Engineering Expected Graduation: December 2026

Relevant Coursework: Microcomputer Systems, Digital Systems Design, Hardware Design Verification, Operating Systems, Circuit Theory, Electronics, Database Systems

Projects

ESP32-Based Chemical Detection System

Senior Capstone Project, 2026–Present

- Collaborating with a **six-person engineering team and industry sponsor** to develop an ESP32-based embedded system for detecting hazardous chemical materials.
- Contributing to a **system-level hardware and firmware architecture** that integrates sensor acquisition, ESP32 data processing, and communication between system components.
- Supporting **sensor integration, calibration planning, verification testing, and prototype validation** through documented requirements and sponsor design reviews.

Arduino UNO R4 WiFi Closed-Loop Drawer Controller [\[GitHub\]](#)

Independent Project, 2026

- Designed an upgraded **Arduino UNO R4 WiFi control architecture** using dual quadrature encoders and four end-of-travel limit switches for position feedback and boundary detection.
- Developed **C++ closed-loop positioning logic** to command each drawer to 0%, 50%, or 100% based on encoder counts instead of fixed motor run times.
- Calibrated full-travel encoder counts and midpoint targets, then bench-tested motor direction, limit-switch cutoff, safety timeouts.

Raspberry Pi 4 Automated Drawer System [\[GitHub\]](#)

Independent Project, 2025

- Built a **GT2 belt-and-pulley drive system** with dual DC motors and **IBT-2 motor drivers** to automate independent drawer movement.
- Developed **Python GPIO/PWM control software** using calibrated 1.4 s and 2.8 s timing intervals to command 50% and 100% drawer travel.
- Implemented **simultaneous dual-motor control** with multi-click push-button commands, allowing both drawers to operate independently without blocking each other.

Skills

Languages: C, C++, Python, Assembly, Verilog, SystemVerilog

Embedded Platforms: ESP32, Arduino UNO R4 WiFi, Raspberry Pi 4

Tools: Arduino IDE, Git, Linux, GDB, JTAG, Vivado, ModelSim/QuartaSim

Technical: GPIO, PWM, sensor/actuator integration, quadrature encoders, limit switches, motor control, closed-loop positioning, hardware debugging, calibration, and validation

Experience

Texas A&M University - College Station

August 2022 – Present

Junior Facilities Coordinator

College Station, TX

- Supported facilities operations by inspecting maintenance work orders, coordinating contractor follow-up, and communicating with BAS/facilities personnel on building system issues.
- Verified completion of contractor and internal maintenance work using Aggie Works DB, documenting deficiencies and initiating follow-up work orders when additional service was required.